

Double Vision

(Diplopia)

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Double vision is seeing 2 images of one object. Double vision may occur when only one eye is open (monocular diplopia) or, more commonly, when both eyes are open (binocular diplopia). Binocular double vision disappears when either eye is closed. Other symptoms, such as eye pain, bulging eye, or muscle weakness, can be present depending on the cause of double vision.

Causes of Double Vision

Monocular double vision can occur when something distorts light transmission through the eye to the retina (the light-sensing structure at the back of the eye). There may be more than 2 images. One of the images is of normal quality (for example, in brightness, contrast, and clarity). The other image or images are of inferior quality.

The most common causes of **monocular double vision** are:

- Clouding of the lens of the eye ([cataract](#))
- Problems with the shape of the cornea (for example, [keratoconus](#), in which the cornea changes from its normal round shape to a conelike shape)
- Uncorrected [refractive error](#), usually astigmatism

Refractive error is imperfect focusing of light rays on the retina. [Astigmatism](#) is refractive error caused by abnormal curvature of the cornea (the clear layer in front of the iris and pupil).

Other causes of monocular double vision include corneal scarring and a dislocated lens.

Binocular double vision suggests the eyes are not pointing at the same object. People normally see an object as a single image even though each eye receives its own separate image of that object. To perceive single images, the eyes must be aligned so that both point to the same object at the same time (called conjugate alignment). When the eyes are not properly aligned, people see 2 images, both of equal quality. Sometimes binocular double vision becomes apparent only when people move their eyes to an extreme in a certain direction (for example, to the far right or left, or up or down).

There are many possible causes of **binocular double vision**. The most common are:

- Paralysis of one of the nerves that controls the muscles that move the eye (the third, fourth, and sixth cranial nerves)
- [Myasthenia gravis](#)
- Mechanical blockage of eye movement

Most commonly, the eyes are misaligned because of a disorder affecting the cranial nerves supplying the muscles that move the eyes, called [extraocular muscles](#). The paralysis may be isolated and the cause may be unknown. Known causes include disorders that typically interfere with the ability of the nerves to control muscles. For example, [myasthenia gravis](#), [botulism](#), and [Guillain-Barré syndrome](#) can affect muscles throughout the body, including the muscles that move the eyes.

Anything that mechanically interferes with eye motion can keep the eyes from aligning properly and cause double vision. Examples include entrapment of an eye muscle in a fracture of the eye socket and deposition of abnormal tissue in the eye socket as can occur with the form of hyperthyroidism called [Graves disease](#).



Some Causes of Double Vision When Both Eyes Are Open

Cause	Common Features*	Diagnostic Approach
Disorders that affect control of eye muscles by the nervous system†		
Certain strokes or transient ischemic attacks	Often in older adults and in people with risk factors for these disorders (such as high blood pressure, atherosclerosis, and diabetes) Sometimes slurred speech, weakness, and/or difficulty walking	A doctor's examination MRI or CT of the brain
A mass that presses on a nerve, such as a bulge in an artery (brain aneurysm) or a tumor	Often pain (sudden if caused by an aneurysm) and often other symptoms of nervous system dysfunction (such as muscle weakness, loss of coordination, and abnormal sensations in the skin)	MRI or CT of the brain
Inflammation or infection of the eye or surrounding structures (for example, abscess, sinusitis, and, rarely, with a blood clot in the cavernous sinus at the base of the skull)	Constant pain Sometimes fever, chills, fatigue, loss of sensation in the face, and/or bulging eyes	CT or MRI of the orbits and sometimes the brain
Multiple sclerosis	Usually periods of relatively good health	MRI of the brain and spinal cord

